Smart Device To Improve Strengthening Exercise For Non-Critical Patient

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I. Introduction

Rehabilitation is essential for individuals who have physical impairments caused by various conditions, helping them work toward restoring their previous level of function. The hands of the human body are vital to everyday life, making them one of the most important body parts. It is also a part of the body that frequently requires rehabilitation due to injuries from accidents or medical issues. Patients such as those with Paresis such as patients recovering after a stroke, Sarcopenia, and Dynapenia are considered to be non-critical patients. These patients are experiencing non-life threatening injuries or diseases however, they are not able to fully exert their strength specifically on their hands. Studies have shown that these patients benefit from progressive resistance training (PRT) [1] [2] [3]

Conventional methods of PRT involve a clinician or a valid professional who will supervise the patient, use tools such as dynamometers to evaluate maximum grip strength and then other resistance tools based on the maximum grip strength value [4]. There is a gap in rehabilitating non-critical patients, the conventional method uses more than one instrument to evaluate and treat patients. Additionally, conventional therapy is time consuming, inaccessible for remote areas, and costly [5].

Telerehabilitation has been a significant solution for a wide range of patients who are not capable of a clinical visit due to cost and accessibility, this method started its trend during COVID-19 pandemic solidifying its importance for home-based approach for stroke survivors [5]. Smart devices are a way to conduct telerehabilitation since they have sensor-integrated equipment, provide real-time feedback, and can track patient progress to help therapists monitor remotely [5].

Commonly used resistance medium of PRTs involve springs and rubber; however through the concept of electromagnetism, fine control of current that changes the magnetic field around an electromagnet can be used as an adjustable repulsion system against an opposite fixed magnetic field. Using this system we can utilize the repulsion between the two magnets as an alternative resistance medium when encased in silicone rubber. The silicone rubber provides a protective barrier against the electrical current. In addition, tactile sensors are placed near the opposing magnets which are used to detect grip strength. Using the data gathered by the tactile sensors, a machine learning algorithm following the guidelines of proper PRT would create a suitable resistance for the patient.

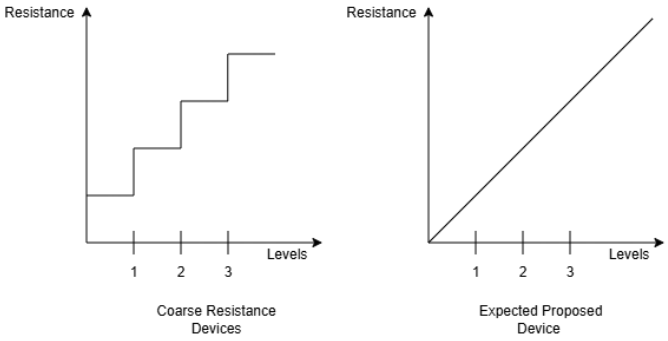


Fig. 1. Comparison of Coarse Resistance Devices and Proposed Device.

Some smart devices lack robust progressive resistance training with only coarse resistance mode such as low medium high resulting in suboptimal recovery however the proposed solution features a fine adjustable resistance control shown on the figure above.

A. Goal and Objectives

The expected output will be a prototype smart recovery gripping tool that can:

- Validate the concept of Electromagnetism as a reliable alternative for resistance control for a rehabilitation tool
- Determine the appropriate stiffness of a silicone rubber for a non-critical patient to conduct gripping exercise
- Accurately collect data with the use of tactile sensors to validate the use of electromagnetism and to monitor a patient recovery progress
- Create a Machine Learning algorithm to provide a suitable PRT for the patient using the device
- A web-based application will be developed to display the data collected from the smart rehabilitation device. The system will store and organize the recorded data to enable continuous monitoring, visualization, and assessment of patient rehabilitation progress over time

B. Scope and Limitation

The study focuses on the design and feasibility to create a smart device that serves as a recovery tool that uses Electromagnetism for resistance adjustment intended for hand rehabilitation for a non-critical patient (Paresis, Sarcopenia, Dynapenia). Tactile sensors are used to capture feedback response to collect data and to aid monitoring the progress. Moreover, the appropriate stiffness of a silicone rubber for the smart device is to be experimentally determined by the flexibility, resistivity from heat, and stiffness of the material. Additionally, the data gathered by the tactile sensors are to be sent in a machine learning algorithm to produce the suitable resistance level of the user.

This study is limited to the assessment of grip strength in hand rehabilitation, including the analysis and monitoring of the hand, fingers, and wrist. The smart device is intended to precisely measure grip force and grip repetitions of non-critical patients performing hand-gripping exercises. A web-based application is utilized for data collection, visualization, and analysis of the rehabilitation data.

C. Significance of the Study

This study aims to develop and test a prototype Smart Grip Rehabilitation Tool that addresses non-critical patient in hand rehabilitation by enabling safe, adjustable resistance training at home or in clinics.

Target Market/Beneficiaries:

- Elderly/mobility-impaired patients (Sarcopenia, Dynapenia).
- Post-injury rehabilitation users needing grip strength tracking
- Stroke survivors with Paresis
- Therapists using remote data monitoring

This research addresses critical gaps in conventional rehabilitation, they are costly and not accessible due to its clinical appointment and availability of professionals leading to a suboptimal recovery. By using tactile sensors that detects force and and controlled electromagnetism, it enables personalized home-based therapy, cutting clinic costs/visits, and an innovative solution.

II. Review of Related Literature

The introduction of using smart devices for rehabilitation has significantly transformed the rehabilitation process for strengthening training for non-critical patients by enabling them to conduct exercises by objective monitoring, adaptive resistance, and data-driven therapy. Traditional rehabilitation methods rely heavily on the presence of a therapist for supervision and subjective evaluation, which may limit the accessibility and progress tracking for patients who are not able to visit for clinical rehab. Recent studies emphasize the use of sensor-based and electromagnetically assisted rehabilitation devices to improve therapy outcomes and provide quantitative feedback [6]

A study published in npj Flexible Electronics introduced a smart-grip strength monitoring device in the form of a sensor-integrated ball using laser-kirigami graphene technology. The device was designed to measure grip force through resistance changes in a deformable graphene strain sensor embedded in a spherical structure [6]. When a user conducts a gripping exercise, the deformation of the ball alters the electrical resistance of the graphene sensors, allowing for a precise measurement of grip strength. The collected data is being transmitted wirelessly to a mobile platform for real-time monitoring and long-term analysis. This approach demonstrates the effectiveness of tactile and force sensors in grip strengthening exercises, particularly for non-critical patients who require low-risk hand rehabilitation and progress tracking outside clinical environment

In contrast to passive monitoring devices, Kim et al. (2022) developed a three-dimensional magnetic finger rehabilitation device aimed at assisting finger movement through controlled electromagnetic forces [7]. The device utilizes electromagnets and permanent magnets attached to a user's fingers to generate attractive and repulsive forces that induce finger flexion, extension, and twisting motions. Unlike conventional motor-driven rehabilitation robots, this system operates without mechanical joints, allowing for a greeted flexibility and ease of use. The pilot study showed that patients who underwent rehabilitation using the magnetic device exhibited improved hand motor function compared to those receiving conventional therapy alone [7]. Although the primary focus of this study was therapeutic effectiveness, the system demonstrates how adjustable electromagnetic resistance can be used to support strengthening exercises in non-critical patients.

Building upon this work, Kim et al. (2023) conducted a larger randomized controlled trial to further evaluate

a magnetic force-driven hand rehabilitation robot. This system incorporated real-time hand position sensing using potentiometers to continuously adjust the direction and magnitude of the magnetic field based on the patient's hand orientation [8]. The adaptive control algorithm ensured that a consistent external force was applied regardless of hand position, enabling reliable and repeatable strengthening exercises. Clinical outcome measures, including the Wolf Motor Function Test and Fugl–Meyer Assessment, indicated significantly greater improvements in the intervention group compared to the control group. This study highlights the importance of combining smart control systems with rehabilitation devices to provide personalized and progressive strengthening exercises

Shore hardness is a standardized empirical measure of a material's resistance to indentation, with values on the Shore A scale ranging from 0 (very soft) to 100 (very hard), where higher values generally indicate stiffer elastomers like rubbers and silicones. The Shore A scale is widely used to characterize softer elastomeric materials, which are particularly relevant for applications requiring flexibility and compliance. In practice, materials selected for therapeutic devices like hand rehabilitation squeeze balls typically fall within the lower to mid-range of the Shore A scale, with softer grades (approximately Shore A 20–40) providing gentle deformation suitable for users with limited grip strength or early rehabilitation stages and medium grades (around Shore A 40–60) offering increased resistance for progressive strength training. These hardness ranges balance flexibility and durability, ensuring user comfort while still allowing effective resistance during repetitive squeezing exercises, making them appropriate for rehabilitation protocols aimed at improving hand strength and function [9].

Sarcopenia is characterized by the age-related loss of skeletal muscle mass and function and dynapenia, defined as the decline in muscle strength independent of muscle mass, significantly impair physical performance and quality of life in older adults. Research has consistently highlighted the role of progressive resistance exercise training (RET) as a primary non-pharmacological intervention for both conditions, as it elicits positive adaptations in the neuromuscular system that increase muscle strength, enhance muscle size, and improve functional capacity in older populations. Specifically, higher intensity and appropriately progressed resistance training protocols have been shown to significantly augment muscle strength and lean body mass, thereby counteracting the muscle atrophy and weakness associated with sarcopenia and dynapenia. This evidence suggests that structured RET programs are effective in both managing and preventing declines in muscle function, with guidelines emphasizing the importance of exercise intensity, volume, and progression to optimize outcomes for older adults [2].

A common feature among the modern smart rehabilitation devices is the integration of data collection and monitoring systems. The graphene-based smart ball focused on collecting grip force and repetition data, which were transmitted wirelessly to a digital platform for analysis [6]. This allows for professionals to assess patient progress objectively and enables patients to perform exercises independently while still being monitored.

Similarly, the magnetic force-driven rehabilitation systems incorporated sensor feedback, such as hand position and movement angles, to adjust therapy parameters in real time [8]. While these systems primarily targeted therapeutic assistance, the recorded sensor data also supported performance evaluation and progress assessment. These findings suggest using tactile sensors with other sensors that detect displacement or deformation of the smart device with a smart control algorithm can enhance both the effectiveness and safety of strengthening exercises for non-critical patients.

III. Methodologies

A. Conceptual Framework

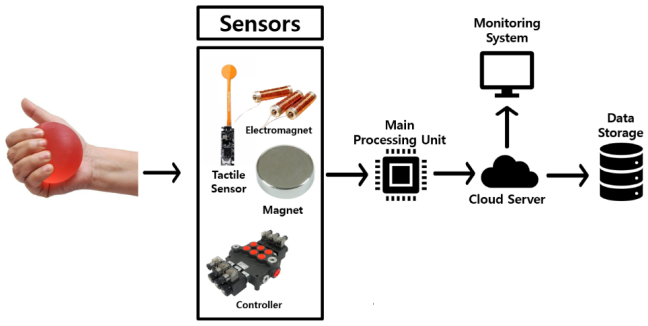


Fig. 2. Conceptual Framework

Grip strength is measured with a single squeeze of the smart grip. The tactile sensor records the pressure, while the magnetic actuator and current controller adjust the resistance. The collected data is sent to a cloud server, stored in a database, and continuously monitored for analysis.

This diagram describes a grip strength measurement and adaptive resistance control process based on tactile sensing and machine learning. The process starts with the user performing a grip action to measure baseline grip strength. During this initial measurement, no current is applied to the electromagnet, and the system only collects tactile sensor data from the finger contact areas.

The measured grip strength data is then sent to the processing unit, where a machine learning model analyzes the input and determines an appropriate resistance level for the user. Based on the model output, an electric current is applied to the electromagnet to generate customized mechanical

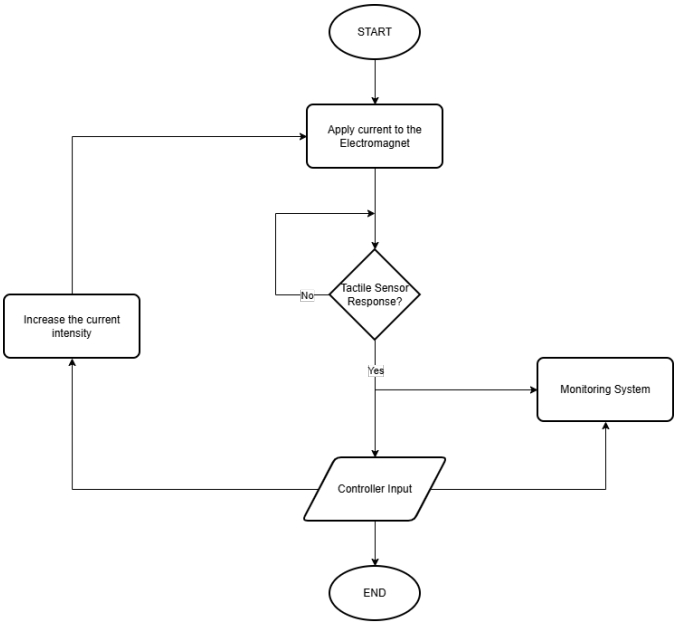


Fig. 3. System Flowchart

resistance. The user then performs the grip exercise under this adjusted resistance.

During the exercise, the system checks whether tactile sensor responses are detected. If the sensor response is insufficient, the controller increases the current supplied to the electromagnet to raise the resistance. If a valid sensor response is detected, the grip force data is recorded and transmitted to the monitoring system for storage and analysis. This closed-loop process enables personalized resistance control and stable grip strength monitoring.

B. System Analysis and Design

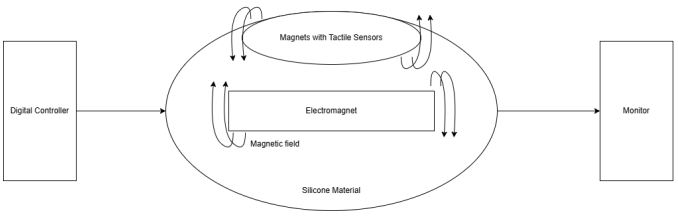


Fig. 4. Device Overview

1.) System Overview

This work proposes a silicone grip tool for repeated grasp training in mild stroke rehabilitation. The body is a silicone ball with finger grooves to keep the hand position consistent across trials. Silicone stiffness is treated as a key design choice because it changes how the grip feels and how pressure reaches the embedded sensors. Several Shore A options are prepared and compared under the same squeeze task. The final Shore A is selected to keep the grip comfortable while still deforming enough to produce clear and stable sensor signals. The groove shape must also keep its form after many repetitions.

2.) Silicone Rubber Experimentation

The initial phase of system development involves experimental evaluation to determine the appropriate Shore A hardness of silicone rubber with respect to flexibility, malleability, and heat resistance. The Shore A scale ranges from 0 to 100, where higher values correspond to increased firmness and greater thermal resistance. The objective of this phase is to identify an optimal Shore A value that provides the lowest possible stiffness suitable for non-critical patients, while still maintaining adequate flexibility, malleability, and heat resistance. Testing begins at the lowest Shore A value and proceeds in increments of five units until the desired material characteristics are achieved.

3.) Electromagnetism Implementation

This phase focuses on identifying the most suitable electromagnet for integration into the system, including an evaluation of solenoid magnets, toroidal magnets, and other applicable electromagnet configurations. Upon selection of the appropriate electromagnet and acquisition of the digital controller, experimental testing is conducted using a hydraulic press machine in conjunction with an ultrasonic sensor to measure displacement. Baseline measurements are first obtained with zero applied current. Subsequently, the current is incrementally increased while maintaining a constant applied pressure, and the resulting changes in system height are recorded. These measurements are used to determine whether the applied electromagnetic field effectively increases the stiffness of the smart device system.

4.) Tactile Sensor Implementation

The tactile sensor module measures pressure changes on the finger contact areas during squeeze and release. Sensors are embedded under the finger-groove surface so the same finger regions press the same spots each time which improves repeatability. The silicone surface acts as a compliant layer that spreads the load and reduces discomfort while still transmitting deformation to the sensing elements.

The sensor signals are sampled continuously during each exercise set and stored with timestamps. A simple filtering step is applied to reduce noise and baseline drift caused by slow silicone recovery. From each squeeze cycle we extract peak level as a force proxy and we also compute cycle-to-cycle variability to represent consistency.

To support reliable use the sensor area is fully encapsulated to block sweat and prevent wire fatigue. The internal layout includes strain relief for cables and a fixed mounting structure so the sensor does not shift after many compressions.

5.) Integration of All Systems The electromagnet module provides adjustable resistance and the tactile module captures how the user responds to that resistance during each squeeze. Resistance is applied through a coil-based magnetic field and is controlled by current commands so the load can be stepped up or down during a session.

During trials the resistance level is logged with timestamps and the tactile signals are sampled in parallel so each repetition has a matched pair of load and pressure features. With higher current settings we expect changes in peak pressure and changes in within-set variability which reflect strength and control.

Electromagnet-related artifacts are also considered in the tactile pipeline. Switching and current changes can introduce electrical noise so grounding and shielding are applied and signal filtering is used to keep the tactile waveform stable. In addition, tactile features and raw sensor streams are transmitted to a monitoring system for real-time visualization and session logging. The monitoring interface displays pressure waveforms and summary metrics alongside the resistance level so clinicians or researchers can review performance during and after training.

C. System Development, Testing and Deployment Strategies

This validation will recruit healthy males aged 20 to 29. Participants will be excluded if they had hand or finger surgery or major injury within the last six months. Those with hand pain or upper-limb musculoskeletal problems will also be excluded. Participants with a history of neurological disorders will be excluded. Written consent will be obtained before participation.

Participants will hold the smart grip with a consistent posture and perform repeated squeeze–release trials. Electromagnetic resistance will be set to three levels. Each level will be tested for a fixed number of repetitions. Tactile signals and resistance settings will be recorded at the same time. After the task, a short questionnaire will assess comfort, slip, grip feel, and perceived difficulty. Success will be defined by stable tactile measurements with limited drift during repeated trials. Tactile features should change consistently when resistance level changes. Usability results should indicate that participants can complete the task without difficulty.

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